

Jiayu Liu

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EDUCATION

Purdue University

West Lafayette, IN, USA

Ph.D. in Technology

2023-2027 (Expected)

Advisor: Dr. Yi Gao and Dr. Volodymyr Bilotkach

Dissertation: A Systematic Framework for Evaluating the U.S. Airports Ground Accessibility and Temporal Reliability Using Gravity-Based Models

M.S. in Applied Statistics

2024-2026

M.S. in Aviation and Aerospace Management

2021-2023

Graduate Certificate in Geospatial Information Science (GIS)

2024

Beijing Jiaotong University

Beijing, China

B.E. in Traffic and Transportation (Civil Aviation Transportation)

2017-2021

CERCIFICATION & LICENSE

- ✧ Part 107 Remote Pilot Certificate
- ✧ SAS Certified Specialist: Base Programming Using SAS 9.4

PEER-REVIEWED PUBLICATIONS

- ✧ Gao, Y., Dou, Z., Wang, S., & Liu, J. (2026). Factors associated with discrepancies between scheduled and actual flight block times: evidence from US domestic airline operations. *Transportation Research Part A: Policy and Practice*, 213, 105227. <https://doi.org/10.1016/j.tra.2026.105227>
- ✧ Liu, J., Kong, N. N., & Gao, Y. (2026). Assessing spatial development footprint of urban areas surrounding new airport terminal infrastructure. *Journal of Air Transport Management*. <https://doi.org/10.1016/j.jairtraman.2025.102957>
- ✧ Gao, Y & Liu, J. (2026). Frequent Flyer Programs in Academic Research: A Literature Review of Research Progress Since 2000. *Journal of Air Transport Management*. <https://doi.org/10.1016/j.jairtraman.2025.102920>
- ✧ Liu, J., & Gao, Y. (Under Review). Mapping Expansion Potential: A Spatial Survey and Land Categorization Analysis for U.S. Airports.
- ✧ Liu, J., Wang, S., Dou, Z., Oluwumi, T. O., & Gao, Y. (2025). Fostering active learning through AI-integrated platforms: Student perceptions from a large enrollment professional education course. *Collegiate Aviation Review International*, 43(2), 69-99. <https://ojs.library.okstate.edu/osu/index.php/CARI/article/view/10322/9131>
- ✧ Liu, J., Wang, S., Dou, Z., & Gao, Y. (2025). Do Aging Aircraft Run Late? Evidence from the United States Domestic Flights. *Journal of the Air Transport Research Society*, 100090. <https://doi.org/10.1016/j.jatrs.2025.100090>
- ✧ Wang, S., Dou, Z., Liu, J., & Gao, Y. (2025). The fate of Akron-Canton Regional Airport following the dehubbing of Cleveland Hopkins International Airport and exodus of Southwest Airlines. *Case Studies on Transport Policy*, 101497. <https://doi.org/10.1016/j.cstp.2025.101497>
- ✧ Liu, J., Beck, C. S., & Gao, Y. (2024). Can you judge a book by its cover? An analysis of travelers' perceptions of airline logos and airlines' business models. *Journal of Air Transport Management*, 115, 102543.

<https://doi.org/10.1016/j.jairtraman.2024.102543>

- ✧ Gao, Y & Liu, J. (2023). State of Airport Loyalty Programs (ACRP Synthesis 131). Washington, DC: *The National Academies Press*. <https://doi.org/10.17226/27279>

CONFERENCE ACTIVITIES

- ✧ Liu, J., & Gao, Y. (2025). Adopting AI-Integrated Online Learning Platforms in Collegiate Aviation Education [Workshop]. *The 2025 UAA Annual Collegiate Aviation Conference & Expo*. Costa Mesa, CA.
- ✧ Liu, J., Kong, N., & Gao, Y. (2024). The role of airport expansions in urban growth: The impact of new terminals on local land use [Poster & Presentation]. *Applied Geography Conference*, San Marcos, TX.
- ✧ Gao, Y. & Liu, J. (2024). Navigating the Skies of Airline Customer Loyalty: A Comprehensive Review of Airline Frequent Flyer Program Studies. *The 27th Air Transport Research Society Conference*, Lisbon, Portugal.
- ✧ Wang, S., Dou, Z., Liu, J., & Gao, Y. (2024). The Ripple Effect of Airport Dehubbing: A case study of Cleveland Hopkins International Airport and Akron Canton Airport [Presentation]. *The 27th Air Transport Research Society Conference 2024*, Lisbon, Portugal.
- ✧ Wang, S., Dou, Z., Liu, J., & Gao, Y. (2023). Spatial Determinants Behind the Seasonal Variations of Commercial Flights Across the United States [Poster & Presentation]. *The 2023 Purdue GIS Day*, West Lafayette, IN.
- ✧ Liu, J., Beck, C., & Gao, Y. (2022). Passengers' Perceptions of Airlines' Logo Design and Their Business Models. *The 25th Air Transport Research Society Conference (Hybrid)*, Antwerp, Belgium.
- ✧ Gao, Y., Liu, J., & Wang, S. (2022). Visualize Your Research Work [Workshop]. *The 2022 UAA Annual Collegiate Aviation Conference & Expo*, Colorado Springs, CO.
- ✧ Wang, S., Liu, J., Xie, H., & Gao, Y. (2022). What is Aviation Research Talking About? A Bibliographic Analysis of Three Aviation Journals during 2011-2021 [Poster & Presentation]. *Purdue University School of Aviation and Transportation Technology Poster Symposium*, West Lafayette, IN.

FUNDED PROJECT

- ✧ **[ACRP] Mapping Expansion Potential: A Spatial Survey for the U.S. Airports** *Principal Investigator* 2025
- ✧ **Sponsor:** Airport Cooperative Research Program (ACRP), Transportation Research Board (TRB), National Academies of Sciences, Engineering, and Medicine (NASEM)
Amount: \$12,000
Period: 11/2025 - 07/2026
Tasks:
 - Define project scope, and coordinate with an industry panel (airport authorities/consultants) to gain practitioner feedback and adoption pathways.
 - Building a GIS-based suitability framework integrating land use, environmental constraints, and FAA infrastructure investment indicators.
 - Producing publishable research paper.

RESEARCH EXPERIENCE & PROJECT

- ✧ **The Data Mine, Purdue University** *Graduate Data Science Researcher* 2025
- ✧ **Corporate Partner:** Johnson & Johnson
Project Scope: Collaborated on Foundational Models project for early-phase clinical trials, applying Generative AI (LLMs, GANs, VAEs, Diffusion Models) and predictive modeling (XGBoost). Applied tools including Python, R, and SQL in large-scale biomedical data analysis.
Tasks:
 - Building data pipelines in Python for large, de-identified biomedical datasets; performing feature selection,

cross-validation, and survival analysis.

- Delivering reproducible notebooks, documentation, and technical reports; collaborating with a cross-functional team on review cycles and presentation of findings.

✧ **[ACRP Synthesis 11-03/Topic S01-27] State of Airport Loyalty Programs** *Lead Research Assistant* 2022

✧ **Sponsor:** Airport Cooperative Research Program (ACRP), Transportation Research Board (TRB), National Academies of Sciences, Engineering, and Medicine (NASEM)

Project Scope: Investigating customer loyalty programs in different domains, especially major airports in the US. Providing a comprehensive overview of the current status of airport loyalty programs.

Tasks:

- Conducting systematic literature review and industry scan on airport loyalty practices; compiling comparative tables of program structures and incentives.
- Assisting with survey design and outreach to airports; cleaning and summarizing responses; drafting synthesis sections on trends, gaps, and implementation challenges.
- Supporting figure/table production and citation management; coordinating edits with PI/co-authors to align with TRB/NAS editorial standards.

✧ **[ACRP 01-41] Airport Blockchain Implementation Guidebook** *Graduate Research Assistant* 2022

✧ **Sponsor:** Airport Cooperative Research Program (ACRP), Transportation Research Board (TRB), National Academies of Sciences, Engineering, and Medicine (NASEM)

Project Scope: Collecting and summarizing blockchain and associated technologies implementation in the aviation industry to provide business solutions and operational efficiencies for airport decision-makers and stakeholders.

Tasks:

- Conducting systematic online scan of blockchain use cases in aviation contexts, including identity and credentialing, asset tracking, maintenance records, supply chain, and payments.
- Produced synthesized summaries, decision checklists, and implementation roadmaps.

TEACHING EXPERIENCE

✧ **AT 102 - Aviation Business**, Purdue University *2024 - 2027*

Role: Graduate Course Instructor

- Working as the instructor for this course, class size approximately 70-150 students vary by semester.
- Solely responsible for designing the syllabus, developing and delivering in-person lectures, creating assessments (quizzes and exams) with clear rubrics, and managing all grading.
- Mentoring student groups on their collaborative projects, guiding topic selection, content organization, and presentation design.
- Integrating current industry insights and inviting guest speakers to connect foundational business concepts with practical applications in the aviation sector.

✧ **AT 546 - Aviation Financial Instruments and Operations**, Purdue University *2021*

Role: Graduate Teaching Assistant

- Managed the airline simulation platform back-end (scenario setup, data resets, system checks), recorded and curated simulation data outputs (KPIs such as load factor, CASM/RASM, on-time performance) for instructor dashboards and student debriefs.
- Supported students with technical troubleshooting, analytics tips, and interpretation of simulation results; graded assignments and short reports with rubric-based feedback.

PROFESSIONAL SERVICE & ACTIVITIES

✧ **Reviewer**

– American Journal of Aerospace Engineering (AJAE)

– Journal of Air Transport Management (JATM)

- Collegiate Aviation Review-International (CAR-i)
 - Journal of Transport Geography (JTG)
 - International Journal of Aviation, Aeronautics, and Aerospace (IJAAA)
 - Transportation Research Board (TRB) Annual Meeting
 - Reliability Engineering & System Safety (RESS)
- ✧ **Professional Affiliation**
- American Association of Airport Executives (AAAE)
 - American Association of Geographers (AAG)
 - Air Transport Research Society (ATRS)
 - University Aviation Association (UAA)

INTERNSHIP

- ✧ **China Mining Logistics Holdings Co., Ltd.** *International Logistics Analyst* **2020**
 – Assisted with freight forwarding, customs declaration and inspection, filling out, photocopying and binding various documents
- ✧ **Citic-Prudential Fund Management Co., Ltd.** *Financial Data Analyst* **2019**
 – Organizing the records of the brokerage roadshow/fund research
 – Collecting and analyzing information on the macroeconomy, capital market, and public offering fund
 – Cleaning up and processing data

SKILLS

- ✧ Statistical Programming/Markup Languages: R, Python, SAS, SPSS, LATEX
- ✧ Languages: English, Mandarin
- ✧ Applications: Microsoft Office, Tableau, ArcGIS Pro, AutoCAD, VOSviewer, Adobe Creative Suite, Premiere Pro
- ✧ Music: Violin Performing and Teaching, Piano, Orchestra Performing

HONORS & AWARDS

- ✧ Graduate Ross Fellowship, Purdue University **2023-2027**
- ✧ Industry Advisory Board Research Poster Symposium **2022**
- ✧ First Prize for 16th National Competition of Transport Science and Technology for University Student **2021**
- ✧ Purdue Bands & Orchestras Scholarship **2021**
- ✧ China Undergraduate Mathematical Contest in Modeling **2020**

COMMUNITY SERVICES & ACTIVITIES

- ✧ **Aviation Graduate Council (AGC), Purdue University** *Member* **Fall 2022**
 – Present for research roundtable
- ✧ **Concert Orchestra, Purdue University** *Concertmaster* **Fall 2022**
 – Organize orchestra rehearsals and concerts
 – First violin sectional rehearsal leader
 – Perform violin solo part
- ✧ **Festival Chorus and Orchestra, Purdue & Ball State University** *Concertmaster* **Summer 2022**
 – Organize rehearsals during summer tour to Europe
 – Perform violin solo part
- ✧ **Philharmonic Orchestra, Purdue University** *Associate Concertmaster* **2021**
 – Perform part of the violin solo works
- ✧ **Suzuki Association of the Americas** *Registered Suzuki Violin Teacher* **2022**
 – Core unites training